

PUSHPENDRA RAGHAV

Research Engineer

Department of Civil, Construction, and Environmental Engineering
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📌 PROFILE HIGHLIGHTS

- **Research Focus:** Integrated hydrologist with expertise in modeling hydrologic processes across the continuum of subsurface, surface, plant, and atmosphere. Research spans four major themes:
 1. Improving water flux predictions (evapotranspiration and soil moisture dynamics) in large-scale land surface models,
 2. Enhancing predictions of streamflow extremes,
 3. Enhancing numerical weather predictions,
 4. Developing stakeholder-relevant tools for early warning of hydrological extremes (such as droughts and floods).
- **Experience:** Over six years of research in eco-hydrology and climatology, including positions as:
 - Research Engineer (Dec. 2024-present), Postdoctoral Associate (Dec. 2023-Dec. 2024), and Graduate Research Assistant (0.5 FTE, Aug. 2019-Dec. 2023), The University of Alabama
 - Junior Research Fellow, Indian Institute of Technology Bombay, India (2018–2019)
 - Graduate Teaching Assistant (0.5 FTE), Indian Institute of Technology Bombay, India (2016–2018)
- **Publications:** Author of 15+ peer-reviewed journal papers (several first-author), in top journals including *Water Resources Research*, *Agriculture and Forest Meteorology*, *Geophysical Research Letters*, *Climate Dynamics*, and *Environmental Research Letters*. Additionally, several manuscripts (8+) currently under review.
- **Awards & Fellowships:** Recipient of Outstanding Dissertation Award (UA, 2025). Ministry of Human Resources Development (MHRD, India) scholarship (2016-18). AGU Chapman Travel Grant.
- **Invention:** Co-inventor of a novel method leveraging potential water-use efficiency vs. evapotranspiration relation to correct eddy covariance latent heat fluxes.
- **Grants & Proposals:** Contributor to multiple external proposals, including a DoE submission on machine learning and land surface modeling (2025). AGU Chapman Travel Grant.
- **Teaching & Mentorship:** Instructor and lecturer for undergraduate and graduate hydrology courses. Mentored graduate students, contributing to student-led publications and fellowship applications. Experience supervising laboratory classes with 120+ students.
- **Professional Service:** Served as reviewer for leading journals including *Water Resources Research*, *Journal of Hydrology*, *JGR–Biogeosciences*, *Remote Sensing of Environment*, and *JGR–Atmospheres*, evaluating the work of other researchers.
- **Judging Activities:** Organized sessions, chaired meetings, and contributed to scientific conferences and workshops. Judge for presentations and posters of PhD and MS students at premier scientific organization of the field, AGU. Served as *facilitator and recorder* for breakout sessions at the AGU Chapman Conference and *co-organizer* of the AmeriFlux Annual Meeting breakout session.
- **Invited Talks:** Delivered multiple invited presentations, including **invited talks at the conference meetings and seminars at universities and research institutes**, reflecting recognition of expertise by both the international research community and academic institutions.
- **Technical Expertise:** Strong background in Python, R, Fortran, and HPC computing. Skilled in ecohydrologic modeling, Earth System Models (ELM, CLM, Ho8, VIC), remote sensing analysis, and machine learning applications for big data in hydrology.

ABOUT

I am a researcher engineer/scientist at the University of Alabama's Department of Civil, Construction, and Environmental Engineering. My work improves understanding and prediction of ecohydrological processes (stream-flow extremes, drought onset, evapotranspiration & soil moisture specifically) through synergistic use of data (including eddy covariance flux data and remote sensing data) and models (including earth system models and data-driven machine learning models).

APPOINTMENTS

Research Engineer , The University of Alabama, USA Department of Civil, Construction, & Environmental Engineering	<i>Dec. 2024-Present</i>
Postdoctoral Associate , The University of Alabama, USA Department of Civil, Construction, & Environmental Engineering	<i>Dec. 2023-Dec. 2024</i>
Graduate Research Assistant , The University of Alabama, USA Department of Civil, Construction, & Environmental Engineering	<i>Aug. 2019-Dec. 2023</i>
Junior Research Fellow , Indian Institute of Technology Bombay, India Climate Studies, Department of Civil Engineering	<i>June 2018-July 2019</i>
Graduate Teaching Assistant , Indian Institute of Technology Bombay Water Resources, Department of Civil Engineering	<i>July 2016-June 2018</i>

EDUCATION

Ph.D. , Civil Engineering (Ecohydrology) The University of Alabama, Tuscaloosa, USA Dissertation: <i>Obtaining Improved Estimates of Soil Moisture, Evapotranspiration, and its partitions using Physical Modeling and Big Data Analytics</i> Advisor: Mukesh Kumar	<i>Aug. 2019-Dec. 2023</i>
M.Tech. , Civil Engineering (Water Resources Engineering) Indian Institute of Technology Bombay, India Thesis: <i>Investigations on the Hydrological Impacts of Climate Change on River Basin Scale using Ho8 Hydrological Model</i> Advisor: T.I. Eldho	<i>July 2016-June 2018</i>
B.Tech. , Civil Engineering Dr. A.P.J. Abdul Kalam Technical University, Lucknow, India	<i>July 2012-June 2016</i>

RESEARCH INTERESTS

Land-atmosphere interactions · Ecohydrology · Remote sensing of hydrosphere and biosphere · Land surface modeling · Big data and machine learning applications · Physics-informed machine learning · Food-Water-Energy Nexus

AWARDS & HONORS

Travel Grant, AGU Energy Balance Chapman, Boulder, CO, U.S.A.	<i>Sep 2025</i>
Outstanding Dissertation Award. Department of Civil, Construction and Environmental Engineering, The University of Alabama. Tuscaloosa, AL.	<i>April 2025</i>
Attained the top rank in the 2016-18 M.Tech Water Resources Engineering batch of the Department of Civil Engineering at Indian Institute of Technology Bombay	<i>June 2018</i>
Ministry of Human Resources Development (MHRD, India) scholarship (2016-18)	<i>July 2016</i>
Scored 99.3 percentile in Graduate Aptitude Test in Engineering (GATE-2016)	<i>2016</i>

INVITED TALKS

Advancing Measurements and Modeling of Invisible Components of Water Budget: Challenges, Opportunities, and Applications, **American Geophysical Union, New Orleans, Louisiana, USA**, Dec. 2025

Flux partitioning in wetland systems, **AmeriFlux Annual Meeting**, Oct. 2025

How Representative Are Ecosystem-Level Evaporative Stress Indices of Vegetation Evaporative Stress?, **American Geophysical Union, Washington, D.C, USA**, Dec. 2024

Evapotranspiration Predictions: Gaps, Challenges, Implications, and the Way Forward, **University of Toledo, Ohio, USA**, Oct. 2024

Model Performance Diagnostics using Information Theory, **RUBISCO Soil Moisture Working Group - Benchmarking**, Aug. 2024

Retrieving Gap-Free Daily Root Zone Soil Moisture Using Surface Flux Equilibrium Theory, **Ohio State University, Columbus, USA**, June 2024

Assessing the impacts of climate change on crop yield in Upper Godavari River Basin using Ho8 hydrological model, **The Inter-Sectoral Impact Model Intercomparison Project (ISIMIP), Paris, France**, June 2019

Parallel Surrogate Global Optimization for Calibrating Multilayer Community Land Model (CLM-MLv5), **Ohio State University, Columbus, USA**, June 2024

RESEARCH GRANTS

VIEW-2: The Virtual Institute for Earth's Water 2026
Proposal Title: FLARE: Field scale retrieval of plant water use And ecosystem RESilience from space
Status: EOI Submitted

Earth and Environmental Systems Science Research in the Southeast U.S. | DEPARTMENT OF ENERGY 2026
FOA Number: DE-FOA-0003600
Proposal Title: Machine-Learning-Enhanced Energy and Water Partitioning in the E3SM Land Model over the Southeastern US: Implications for Energy Systems-Relevant Environmental Variables
Program Office: Biological and Environmental Research (BER) program within the Department of Energy (DOE)'s office of Science
Status: Under Review

ARS AI Innovation Fund | U.S. DEPARTMENT OF AGRICULTURE 2026
Proposal Title: Developing a High-Resolution Root-Zone Soil Moisture Product over CONUS via Surface Flux Equilibrium Theory-Informed Machine Learning
Program Office: ARS Artificial Intelligence Center of Excellence (AI-COE) within the U.S. Department of Agriculture (USDA)
Status: Under Review

AGU Chapman Travel Grant 2025
Proposal Title: A Novel Water Use Efficiency-Based Method for Correcting Eddy Covariance Latent Heat Fluxes for Energy Imbalance
Status: Approved

Future Investigators in NASA Earth and Space Science and Technology (FINESST) 2022
Proposal Number: 21-EARTH21-0320
Proposal Title: Assessment of evapotranspiration partitioning estimates from ECOSTRESS
Overall Score: Very Good
Status: Declined

Alabama EPSCoR Graduate Research Scholarship 2020
Proposal Title: Fine Resolution Mapping of Evapotranspiration from Rain-fed and Irrigation-fed Fields in Alabama
Status: Declined

PUBLICATIONS & CONFERENCES

* Authors contributed equally

 Google Scholar

 ORCID

 ResearchGate

PEER-REVIEWED JOURNAL PAPERS

(Published)

Raghav, P., & Kumar, M. (2026). PULSE: A Novel Potential Underlying Water Use Efficiency-Based Method for Latent Heat and Surface Energy Imbalance Correction. *Water Resources Research*. <https://doi.org/10.1029/2025WR042766>

Raghav, P., & Kumar, M. (2026). Dynamic Parameterization of Priestley-Taylor Coefficient for a More Accurate Potential Evapotranspiration Estimation in Ungauged Regions. *Environmental Research Letters*. <https://doi.org/10.1088/1748-9326/ae6464>

Massoud, E. C., Collier, N., Wang, Y., Mao, J., Harpold, A., Kannenberg, S. A., Koren, G., Kumar, M., Raghav, P., Ray, P.,

- Shi, M., Tao, J., Vasu, S. P., Wang, H., Zhu, Q., & Hoffman, F. M. (2026): Benchmarking soil moisture and its relationship to ecohydrologic variables in Earth System Models. *Geoscientific Model Development*. <https://doi.org/10.5194/gmd-19-3427-2026>
- Salvi, K., Raghav, P., Kumar, M., Magliocca, N., & Bisht, G. (2026). Discrepancy in the Sign of Temperature Trends in Reanalysis Datasets. *Environmental Research Letters*. <https://doi.org/10.1088/1748-9326/ae5726>
- Thakur, H., Raghav, P., & Kumar, M. (2026). How well does the Evaporative Stress Index from ECOSTRESS Capture Site-Based Stresses? *Canadian Journal of Remote Sensing*, 52(1). <https://doi.org/10.1080/07038992.2026.2645742>
- Thakur, H., Raghav, P., & Kumar, M. (2025). Surface Flux Equilibrium Theory-Derived Evapotranspiration Estimate Outperforms ECOSTRESS, MODIS, and SSEBop products. *Geophysical Research Letters*, 52(10), e2025GL114822.. <https://doi.org/10.1029/2025GL114822>
- Ghaseminejad, A., Bhat, N., Raghav, P., & Kumar, M. (2025). Influence of Interannual Leaf Phenology Dynamics on Evapotranspiration Predictions. *Journal of Hydrometeorology* 26, 259–272. <https://doi.org/10.1175/JHM-D-24-0061.1>
- Raghav, P., & Kumar, M. (2024). Are the ecosystem-level evaporative stress indices representative of evaporative stress of vegetation?. *Agricultural and Forest Meteorology*, 357, 110195. <https://doi.org/10.1016/j.agrformet.2024.110195>
- Raghav, P., Kumar, M., & Liu, Y. (2024). Structural constraints in current stomatal conductance models preclude accurate prediction of evapotranspiration. *Water Resources Research*, 60(8), e2024WR037652. <https://doi.org/10.1029/2024WR037652>
- Rathore, L. S., Hanasaki, N., Kumar, M., Mekonnen, M., & Raghav, P. (2023). Water scarcity challenges across urban regions with expanding irrigation. *Environmental Research Letters*, 19(1), p.014065. <https://doi.org/10.1088/1748-9326/ad178a>
- Raghav, P., & Eldho, T.I. (2023). Investigations on the hydrological impacts of climate change on a river basin using macroscale model Ho8. *Journal of Earth System Science*, 132(2), p.87. <https://doi.org/10.1007/s12040-023-02102-4>
- Wagle, P.*, Raghav, P.*, Kumar, M., & Gunter, A.G. (2022). Influence of water use efficiency parameterizations on flux variance similarity-based partitioning of evapotranspiration. *Agricultural and Forest Meteorology*, 328, p.109254. <https://doi.org/10.1016/j.agrformet.2022.109254>
- Raghav, P., Wagle, P., Kumar, M., Banerjee, T., & Neel, J.P.S (2022). Vegetation Index-Based Partitioning of Evapotranspiration Is Deficient in Grazed Systems. *Water Resources Research*, 58(8), p.e2022WR032067. <https://doi.org/10.1029/2022WR032067>
- Raghav, P., & Kumar, M. (2021) “Retrieving gap-free daily root zone soil moisture using surface flux equilibrium theory. *Environmental Research Letters*, 16(10), p.104007. <https://doi.org/10.1088/1748-9326/ac2441>
- Raghav, P., Borkotoky, S. S., Joseph, J., Chattopadhyay, R., Sahai, A. K., & Ghosh, S. (2020). Revamping extended range forecast of Indian summer monsoon. *Climate Dynamics*, 55, pp.3397–3411. <https://doi.org/10.1007/s00382-020-05454-5> (In revision; drafts are available upon request)
- Raghav, P., Liu, Y., Kumar, M., & Bisht, G. (2026). Multilayer canopy model outperforms big-leaf model for evapotranspiration predictions under high water and heat stress conditions. *Journal of Advances in Modeling Earth Systems* (In Review).
- Raghav, P., & Kumar, M. (2026). Padé Approximation of the Clausius-Clapeyron Relation: An Explicit and Computationally Efficient Alternative to the Penman-Monteith Evapotranspiration Equation. *Geophysical Research Letters* (In Review).
- Thakur, H., Raghav, P., & Kumar, M. (2026). Benchmark data of root zone soil moisture over the conterminous USA based on vegetation-specific rooting depths. *Vadose Zone Journal* (In Review).
- Wolkeba, F. T., Kumar, M., Raghav, P., & Thakur, H. (2026). CALFRET: A high-performing, CALibration-FRee 1-km Evapotranspiration product over the conterminous USA. *ESS Open Archive*. 05 January 2026. [preprint], <https://doi.org/10.22541/essoar.175915560.04888321/v3> (Water Resources Research (In Revision)).
- Wagle, P.*, Raghav, P., Kumar, M., Scanlon, T.M., Northup, B.K., Moffet, C., Gunter, S., & Xiao, X. Partitioning Evapotranspiration in C3-C4 Mixed Tallgrass Prairies: Comparing Year-round C4 Parameterization with Dynamic Seasonal C3-C4 Parameterization. *Agricultural Water Management* (In Review).
- Moreno, H., Massoud, E., Yaari, A., Huang, J., Koren, G., Raghav, P., et al. (2026). How is Artificial Intelligence Enhancing Soil Moisture Estimation and Prediction? *Draft in preparation*.
- Desai, A., Wanner, L., Levy, P., Zhang, W., Durden, D., Metzger, S., Raghav, P., et al. (2026). Evaluating mechanistic corrections for surface energy imbalance in eddy covariance. *In preparation*.
- Raghav, P., & Kumar, M. (2026). Large Underestimation of Flash Drought Events in Reanalysis Data. *Draft in preparation for Geophysical Research Letters*.

Raghav, P., & Kumar, M. (2026). Evapotranspiration dynamics and partitioning in soybean under different irrigation practices. *Draft in preparation for Agricultural and Forest Meteorology*.

Raghav, P., & Forbrich, I. (2026). Evapotranspiration partitioning in tidal wetlands across hydrological, salinity, and vegetation gradients. *In preparation*.

■ BOOK CHAPTER

Raghav, P., & Eldho, T.I. (2021). Assessing the Impacts of Climate Change on Crop Yield in Upper Godavari River Subbasin Using Ho8 Hydrological Model. *Climate Change Impacts on Water Resources: Hydraulics, Water Resources and Coastal Engineering*, 193-205. https://doi.org/10.1007/978-3-030-64202-0_18

🎤 CONFERENCE PRESENTATIONS

* Presenting author/convenor

Raghav, P.*, & Kumar, M. (2025). Evaluating the Accuracy of Evaporative Stress Index-based Flash Drought Detection in Global Reanalysis Data. *American Geophysical Union Fall Meeting*: H21B-04, New Orleans, LA.

Raghav, P., & Kumar, M.* (2025). Leveraging FluxNet Data for Improved Estimation of Potential Evapotranspiration Using the Priestley-Taylor Method. *American Geophysical Union Fall Meeting*: B12C-07, New Orleans, LA.

Salvi, K., Raghav, P.*, & Kumar, M. (2025). Evaluation of Temperature Trends Simulated by Reanalysis Data at Observations Sites in the Continental United States. *American Geophysical Union Fall Meeting*: BH11M-1043, New Orleans, LA.

Kumar, M.*, Raghav, P., Liu, Y., & Bisht, G. (2025). The Complexity Paradox: When Advanced Evapotranspiration Models Underperform (invited). *American Geophysical Union Fall Meeting*: H52A-02, New Orleans, LA.

Nishan, B.*, Wagle, P., Raghav, P., & Shayeghi, A. (2025). Advancing Measurements and Modeling of Invisible Components of the Water Budget: Challenges, Opportunities, and Applications | Oral. *American Geophysical Union Fall Meeting*: H13M, New Orleans, LA.

Tang, A.*, Raghav, P.*, & Forbrich, I. (2025). Partitioning Evapotranspiration in Coastal Wetlands Using Flux Variance Similarity: Multi-Year Insights into Drought, Hydrology, and Vegetation | Online. *American Geophysical Union Fall Meeting*: H42D-06, New Orleans, LA.

Wolkeba, F*, Kumar, M., Raghav, P., & Thakur, H. (2025). A High-Performing 1 km Daily Gridded Evapotranspiration Dataset for CONUS Based on Surface Flux Equilibrium Theory. *American Geophysical Union Fall Meeting*: H41D-07, New Orleans, LA.

Raghav, P.*, & Kumar, M. (2025). Evaluating the Influence of Model Parameters, Structure, Input Data, and Site-Specific Characteristics on Energy Partitioning Errors in Land Surface Models. *In Chapman Conference on the Energy Balance Closure Problem*. AGU, Sep 14-19, 2025 in Boulder, CO.

Raghav, P.*, Liu, Y., & Kumar, M. (2024). Enhancing Water and Carbon Flux Predictions: A Comparative Study of Multi-Layer Canopy and Big-Leaf Models under Varying Water Stress and Light Conditions. *American Geophysical Union Fall Meeting*, Dec 9-13, 2024 in Washington, D.C.

Raghav, P.*, & Kumar, M. (2024). How Representative Are Ecosystem-Level Evaporative Stress Indices of Vegetation Evaporative Stress? *American Geophysical Union Fall Meeting*, Dec 9-13, 2024 in Washington, D.C.

Raghav, P.*, & Kumar, M. (2024). The accuracy of evapotranspiration prediction is impeded by structural limitations present in current stomatal conductance models. *The Ecological Society of America Annual Meeting*, August 4-9, 2024 in Long Beach, CA.

Raghav, P.*, & Kumar, M. (2023). A physics-informed machine learning approach to study global dynamics of transpiration to evapotranspiration ratio. *American Geophysical Union Fall Meeting*: B43E-2592, San Francisco, CA. <https://ui.adsabs.harvard.edu/abs/2023AGUFM.B43E2592R/abstract>

Raghav, P.*, & Kumar, M. (2023). Evapotranspiration partitioning in evaporation-dominant settings. *American Geophysical Union Fall Meeting*: H21Q-1573, San Francisco, CA. <https://ui.adsabs.harvard.edu/abs/2023AGUFM.H21Q1573R/abstract>

Ghaseminejad, A., Raghav, P., & Kumar, M.* (2023). Is the Representation of Interannual Leaf Phenology Dynamics Needed for Accurate Estimation of Evapotranspiration?. *American Geophysical Union Fall Meeting*: B43E-2604, San Francisco, CA. <https://ui.adsabs.harvard.edu/abs/2023AGUFM.B43E2604K/abstract>

Wagle, P*, Raghav, P., Kumar, M., Scanlon, T.M., Northup, B.K., Moffet, C., Gunter, S., & Xiao, X. (2023). A Comparison of FVS, MREA, and CEC Methods to Partition Evapotranspiration in Tallgrass Prairies. *American Geophysical Union Fall Meeting*: H13D-06, San Francisco, CA. <https://ui.adsabs.harvard.edu/abs/2023AGUFM.H13D..06W/abstract>

Raghav, P.*, & Kumar, M. (2022). Assessing evapotranspiration partitioning obtained using land surface models across varied ecosystems. *American Geophysical Union Fall Meeting*: B12E-1121, Chicago, IL. <https://ui.adsabs.harvard.edu/>

[abs/2022AGUFM.B12E1121R/abstract](https://ui.adsabs.harvard.edu/abs/2022AGUFM.B12E1121R/abstract)

Ghaseminejad, A.*, Bhat, M., **Raghav, P.**, & Kumar, M. (2022). Machine Learning-Driven Temporal Downscaling of Groundwater Recharge Estimates to Obtain Event-Based Response. *American Geophysical Union Fall Meeting*: H35H-1215, Chicago, IL. <https://ui.adsabs.harvard.edu/abs/2022AGUFM.H35H1215G/abstract>

Kumar, M.*, Rathore, L. S., Hanasaki, N., Mekonnen, M., & **Raghav, P.** (2022). Rain-fed to Irrigation-fed Transition of Cropped Agriculture may Enhance Urban-Rural Water Conflict. *American Geophysical Union Fall Meeting*: GC35D-05, Chicago, IL. <https://ui.adsabs.harvard.edu/abs/2022AGUFMGC35D..05K/abstract>

Wagle, P.*, **Raghav, P.**, & Kumar, M. (2022). Sensitivity of Water Use Efficiency Parameterizations on Flux Variance Similarity-based Partitioning of Evapotranspiration. *American Geophysical Union Fall Meeting*: B12E-1123, Chicago, IL. <https://ui.adsabs.harvard.edu/abs/2022AGUFM.B12E1123W/abstract>

Raghav, P.*, & Kumar, M. (2021). Are canonical relations to partition evapotranspiration valid for managed systems?. *American Geophysical Union Fall Meeting*: B12B-02, New Orleans, LA. <https://ui.adsabs.harvard.edu/abs/2021AGUFM.B12B..02R/abstract>

Raghav, P., & Kumar, M.* (2021) Obtaining daily root zone soil moisture using a calibration-free approach. *American Geophysical Union Fall Meeting*: H55C-0777, New Orleans, LA. <https://ui.adsabs.harvard.edu/abs/2021AGUFM.H55C0777K/abstract>

Raghav, P., & Eldho, T.I.* (2019). Investigations on the impacts of future climate change on the hydrology of a river basin in India using a macroscale hydrological model. *11th World Congress on Water Resources and Environment: Managing Water Resources for a Sustainable Future-EWRA*: p. 469-470, Madrid, Spain. http://ewra.net/pages/EWRA2019_Proceedings.pdf

Raghav, P.* (2019). Assessing the impacts of climate change on crop yield in Upper Godavari River Basin using Ho8 hydrological model. *The Inter-Sectoral Impact Model Intercomparison Project (ISIMIP) Cross sectoral workshop*, Paris, France.

THESIS REPORTS

Raghav, P. Obtaining Improved Estimates of Soil Moisture, Evapotranspiration and its Components Using Physical Modeling and Big Data Analysis. **Ph.D. Dissertation**

Supervisor: [Mukesh Kumar](#), Department of Civil Engineering, The University of Alabama, Tuscaloosa, USA, Dec, 2023. ([Link](#))

Raghav, P. Investigations on the Hydrological Impacts of Climate Change on River Basin Scale using Ho8 Hydrological Model. **M.Tech. Thesis**

Supervisor: [T.I. Eldho](#), Department of Civil Engineering, Indian Institute of Technology Bombay, India, June, 2018. ([Link](#))

SOFTWARE & DATASETS

Evaluating multilayer and big-leaf canopy models for evapotranspiration under water and heat stress (Zenodo)	Jan 2026
Dynamic Parameterization of Priestley-Taylor Coefficient for a More Accurate Potential Evapotranspiration Estimation in Ungauged Regions (Zenodo)	Nov 2025
PULSE: Correcting Latent Heat Flux Measurements for Energy Imbalance at Eddy Covariance Flux Tower Sites Based on the Concept of Potential Underlying Water Use Efficiency (Zenodo)	Oct 2025
Water Flux Partitioning using Flux Variance Similarity Theory in Wetland Settings (GitHub)	Oct 2025
Guide to download and process remote sensing data in R (GitHub)	April 2025
Information Theory-based Model performance diagnostics (GitHub)	Aug. 2024
Ecosystem-scale Transpiration from SAPFLUXNET Database (GitHub)	April 2024
Water Flux Partitioning using high frequency EC data (GitHub , Zenodo)	Aug. 2023
Physical and Machine Learning models of stomatal conductance for evapotranspiration prediction (Zenodo)	March 2023
Water Flux Partitioning Data in Disturbed (Grazed) Wheat systems (USDA-ADC)	Aug. 2022
Step-by-step guide for gap filling eddy covariance data and NEE flux partitioning (GitHub)	2022
Evapotranspiration partitioning methods (GitHub)	2022
Extracting Spatial Data from NASA data repository (GitHub)	2022

SOFTWARE EXPERTISE

Earth System Models

Energy Exascale Earth System Model (E3SM) Land Model (ELM) | Community Land Model (CLM) | Ho8 | VIC | SWMM | ILAMB

Programming Languages

Python | R | Matlab | Fortran | LaTeX | Bash

Debugging Tools

GDB

Operating Systems

Linux | Windows

Geospatial analysis / Plotting Tools

PySTAC | GEE | ArcGIS | QGIS | AutoCAD (Civil 3D) | ERDAS Imagine | GMT | Matplotlib | ggplot | OriginPro | folium

Used important packages/libraries

xarray | dask-ml | h5py | rasterio | geopandas | earthengine-api | cartopy | climate4R | Keras/TensorFlow | PyTorch

TEACHING & MENTORSHIP

Graduate Teaching Assistant

Mentored 120 undergraduate students at their time of experiments in hydraulics laboratory along with evaluation and supervision

2016-2018

Indian Institute of Technology Bombay, India

Teaching Process Hydrology (CEE-491/591)

2023-2024

The University of Alabama, USA

Hitesh P. Thakur, Ph.D Student, The University of Alabama (Mentor)

August 2024 to present

PROFESSIONAL SERVICE

JOURNAL REVIEWER

Water Resources Research

JGR-Atmosphere

JGR-Biogeosciences

Remote Sensing of Environment

Journal of Hydrology

Journal of Water and Climate Change

MEMBER & RESEARCH COMMUNITIES

The Ecological Society of America

American Geophysical Union

The Indian Society for Hydraulics

Alabama Water Institute